

Wei Zhang

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SUMMARY

AI/ML researcher and engineering leader with 10+ years of experience in large-scale transformer-based language models, deep learning, and distributed systems. Former Research Scientist at IBM T.J. Watson Research Center and Head of AI at Material Bank. Best paper on human and AI collaboration on top conference CHI 2020. First author or core contributors to AAAI, ICML, NeurIPS, ACL conferences as well as taking roles of program committee members.

Expertise in designing and running large-scale experiments, building data pipelines for model training, and optimizing ML systems from model architecture to infrastructure. Strong background in transformer models, reinforcement learning, and interpretable NLP.

Research interests include language model pre-training, scaling, efficiency, and alignment.

EDUCATION

Master of Science (Language Technologies)

Carnegie Mellon University

2014

Pittsburgh, PA

Bachelor of Science (Computer Science)

Harbin Institute of Technology

2007

China

EXPERIENCE

Co-founder, CTO, Chief AI Scientist

Paladio AI

2025 – Present

USA

- Designed and built large-scale AI systems including LLM-based agents, data pipelines, and distributed inference systems.
- Developed scalable pipelines transforming unstructured data into model-ready datasets (parsing, normalization, entity extraction).
- Built end-to-end ML infrastructure spanning data ingestion, training workflows, and deployment on AWS (ECS, S3).
- Optimized system performance for latency, throughput, and cost efficiency.
- Developed multi-agent orchestration systems with focus on scalability and reliability.

Head of AI and Data Science

Material Bank

2023 – 2025

USA

- Led development of ML and LLM-based systems for large-scale structured and unstructured data.
- Built data pipelines supporting ML training and real-time inference.
- Developed retrieval and ranking systems using deep learning and LLM-based relevance modeling.
- Designed production ML systems handling high-throughput workloads.

Principal Machine Learning Engineer

Saks Fifth Avenue

2022 – 2023

USA

- Developed transformer-based and multi-modal models (e.g., CLIP) for semantic search.
- Designed ranking systems leveraging deep learning and behavioral data.
- Improved performance through architecture and pipeline optimization.

Senior Staff Data Scientist

Wayfair

2021 – 2022

USA

- Led research on semantic search and representation learning.
- Designed large-scale experimentation and A/B testing frameworks.
- Built knowledge graph and embedding systems for retrieval tasks.

Research Scientist

IBM T. J. Watson Research Center

2014 – 2021

USA

- Conducted research on transformer-based language models, neural machine comprehension, and reinforcement learning.
- Designed and ran large-scale experiments on model performance and representation learning.
- Worked on attention-based architectures and model interpretability.
- Published in AAAI, ACL, NeurIPS workshops, CHI (Best Paper Award).
- Developed reinforcement learning methods for multi-agent systems.

SELECTED RESEARCH & SYSTEMS EXPERIENCE

- Language Models: Experimentation and evaluation of transformer-based architectures.
- Data Systems: Design of large-scale ETL pipelines for training data preparation.
- Distributed Systems: Experience with AWS-based scalable infrastructure.
- Model Optimization: Performance tuning across models and systems.
- Reinforcement Learning: Multi-agent and sequential decision-making systems.

TECHNICAL SKILLS

Machine Learning: Deep learning, transformers, NLP, reinforcement learning

Frameworks: PyTorch, TensorFlow (familiar), NumPy, SciPy

Systems: Distributed systems, AWS (ECS, S3), data pipelines, microservices

Languages: Python (expert), Java

HONORS AND AWARDS

CHI Best Paper Award

Human-AI interaction research

2020

ACL SemEval Winner

Word Sense Disambiguation

2007